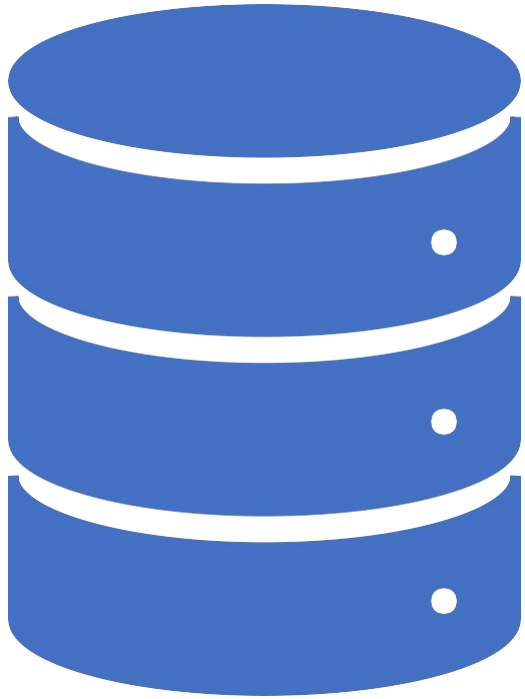
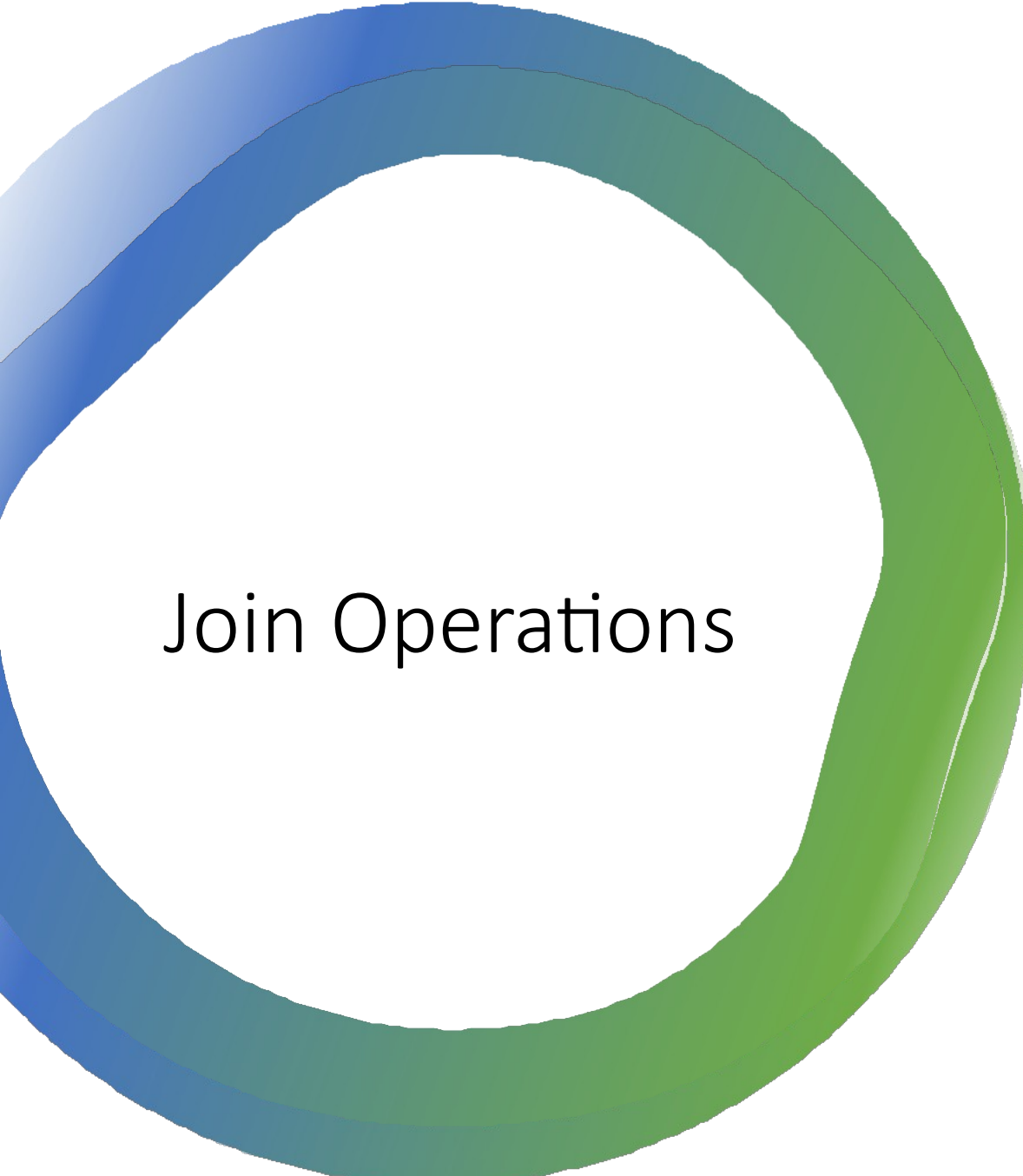




Database System
[Join Operation in DBMS]



Join Operations

- A Join operation ***combines related tuples from different relations, if and only if a given join condition is satisfied.***
- It is denoted by $\bowtie$.



Join Operations Example

- **EMPLOYEE**

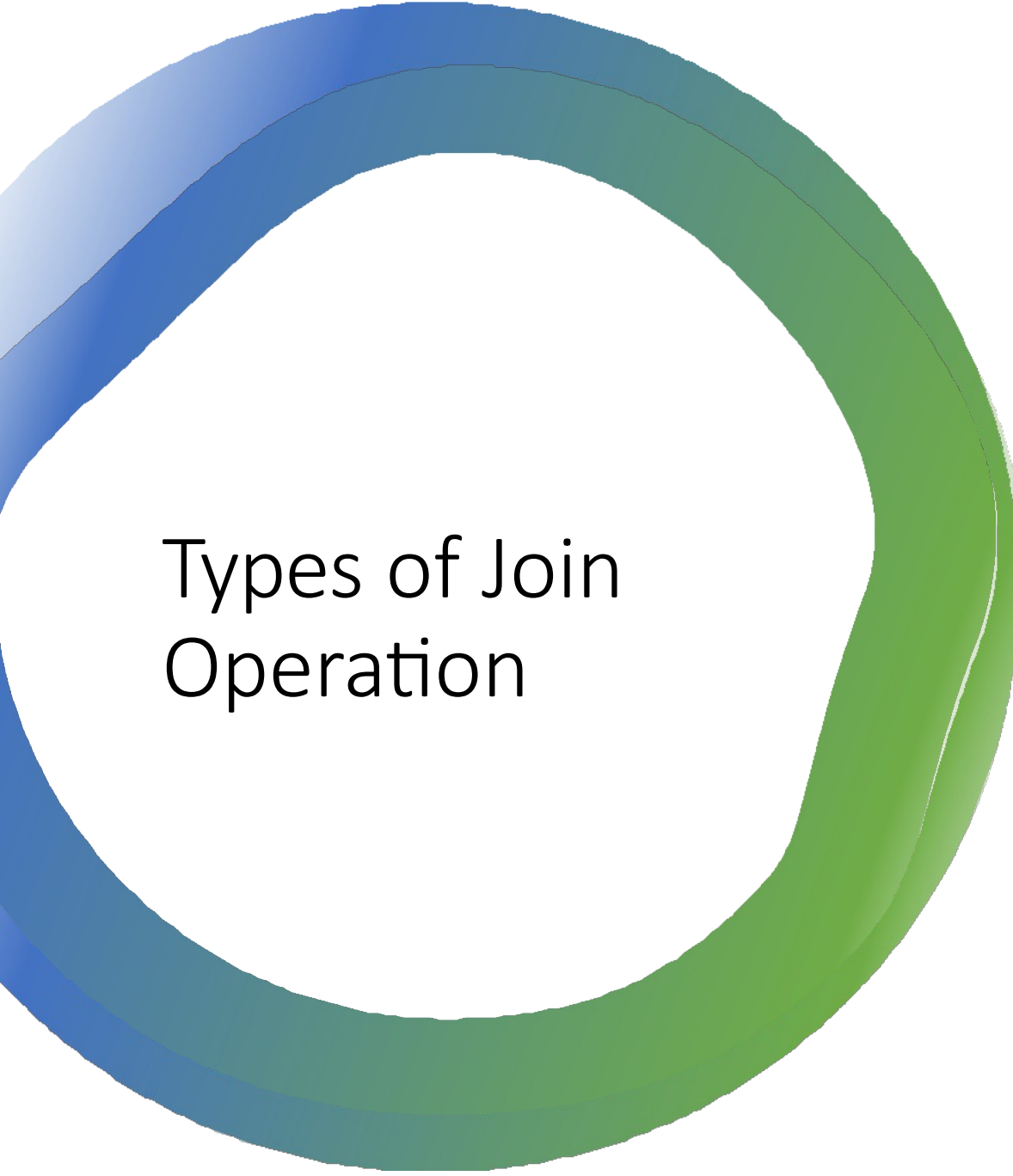
EMP_CODE	EMP_NAME
101	Stephan
102	Jack
103	Harry

- **SALARY**

EMP_CODE	SALARY
101	50000
102	30000
103	25000

Operation: (**EMPLOYEE** ⋈ **SALARY**)

EMP_CODE	EMP_NAME	SALARY
101	Stephan	50000
102	Jack	30000
103	Harry	25000



Types of Join Operation

Natural
Join

Outer Join

Equi Join

Left Outer
Join

Right
Outer Join

Full Outer
Join



Natural Join

- A natural join is the ***set of tuples of all combinations in R and S that are equal on their common attribute names.***
- It is denoted by $\bowtie$.



Natural Join Example

- **EMPLOYEE**

EMP_CODE	EMP_NAME
101	Stephan
102	Jack
103	Harry

- **SALARY**

EMP_CODE	SALARY
101	50000
102	30000
103	25000

INPUT: $\pi_{EMP_NAME, SALARY}(EMPLOYEE \bowtie SALARY)$

OUTPUT:

EMP_NAME	SALARY
Stephan	50000
Jack	30000
Harry	25000



Outer Join

- The outer join operation is an ***extension of the join operation.***
- It is used to ***deal with missing information.***



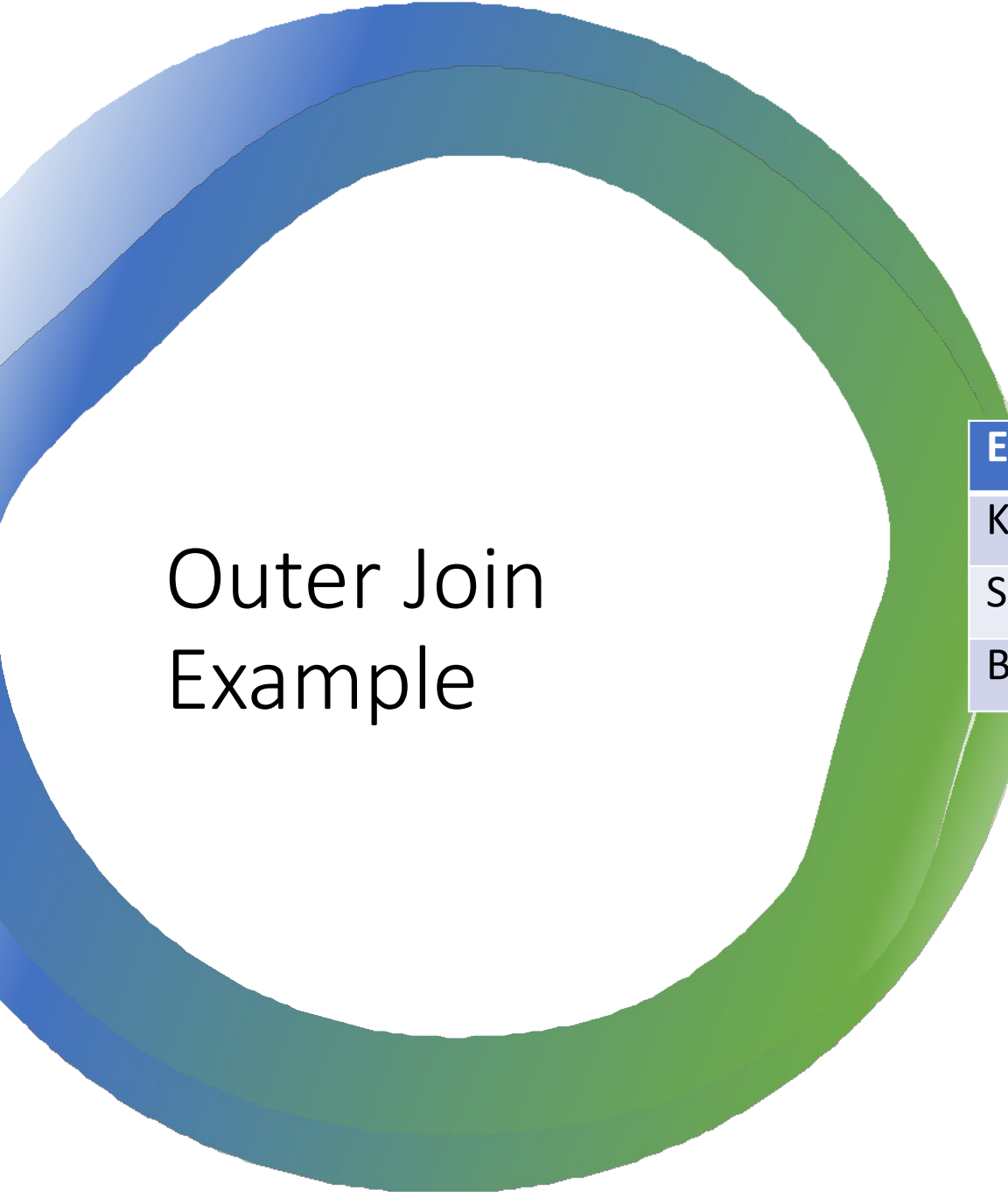
Outer Join Example

• **EMPLOYEE**

EMP_NAME	STREET	CITY
Kamran	Civil line	Muzaffarabad
Saqib	Park street	Kel
Haider	M.G. Street	Lahore
Bilawal	View city	Hyderabad

• **FACT_WORKERS**

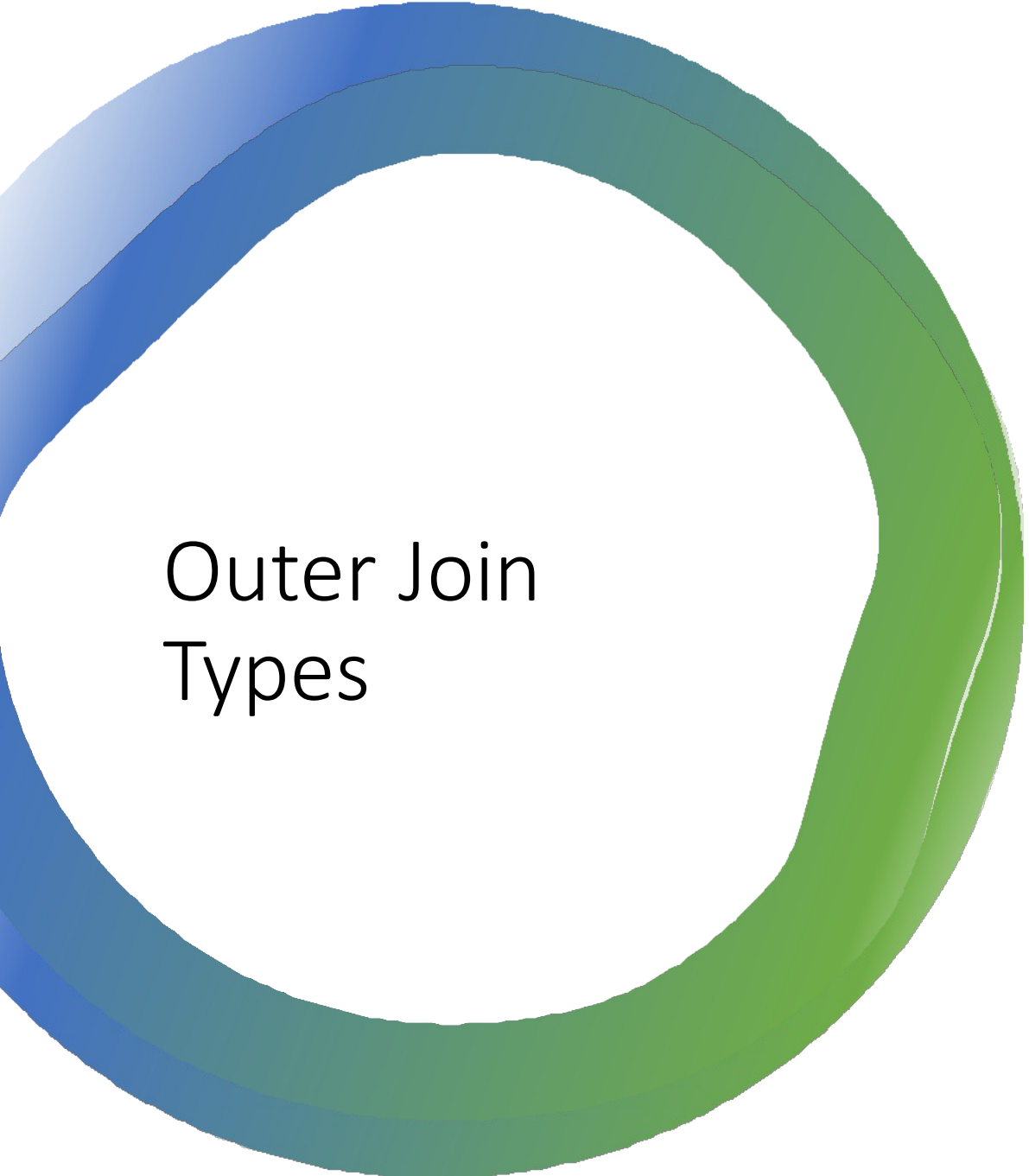
EMP_NAME	BRANCH	SALARY
Kamran	Infosys	10000
Saqib	Wipro	20000
Ayyaz	HCL	30000
Bilawal	TCS	50000



Outer Join Example

- **INPUT:** (EMPLOYEE ⋈ FACT_WORKERS)
- **OUTPUT:**

EMP_NAME	STREET	CITY	BRANCH	SALARY
Kamran	Civil line	Muzaffarabad	Infosys	10000
Saqib	Park street	Kel	Wipro	20000
Bilawal	View city	Hyderabad	TCS	50000



Outer Join Types

- An outer join is basically of three types:
 - Left outer join
 - Right outer join
 - Full outer join



Left Outer Join

- Left outer join contains the ***set of tuples of all combinations in R and S that are equal on their common attribute names.***
- In the left outer join, ***tuples in R have no matching tuples in S.***
- It is denoted by $\bowtie$.



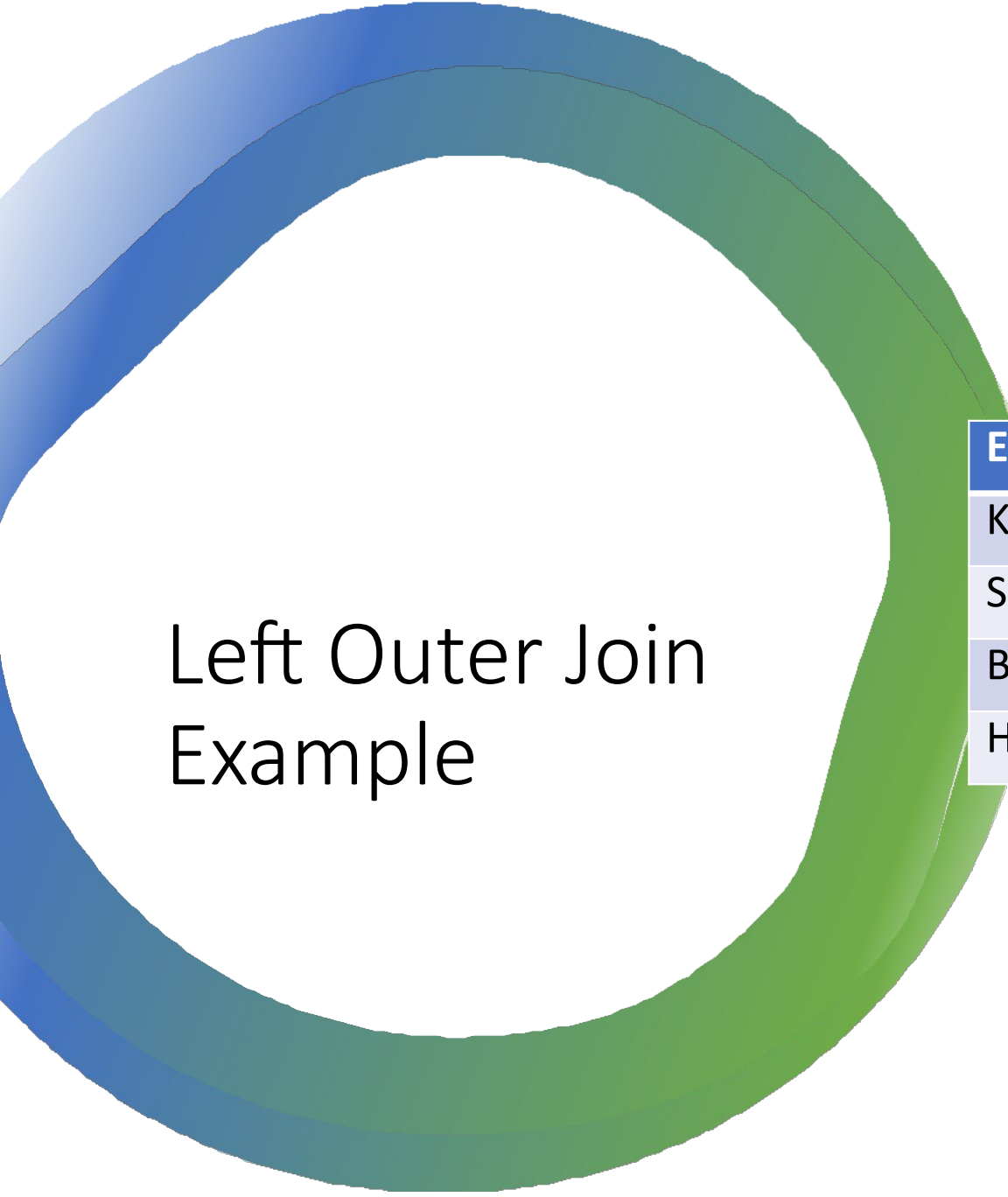
Left Outer Join Example

• EMPLOYEE

EMP_NAME	STREET	CITY
Kamran	Civil line	Muzaffarabad
Saqib	Park street	Kel
Haider	M.G. Street	Lahore
Bilawal	View city	Hyderabad

• FACT_WORKERS

EMP_NAME	BRANCH	SALARY
Kamran	Infosys	10000
Saqib	Wipro	20000
Ayyaz	HCL	30000
Bilawal	TCS	50000



Left Outer Join Example

- **INPUT:** (EMPLOYEE to
FACT_WORKERS)

- **OUTPUT:**

EMP_NAME	STREET	CITY	BRANCH	SALARY
Kamran	Civil line	Muzaffarabad	Infosys	10000
Saqib	Park street	Kel	Wipro	20000
Bilawal	View city	Hyderabad	TCS	50000
Haider	M.G. Street	Lahore	NULL	NULL



Right Outer Join

- Right outer join contains the ***set of tuples of all combinations in R and S that are equal on their common attribute names.***
- In the right outer join, ***tuples in S have no matching tuples in R.***
- It is denoted by $\bowtie$.



Right Outer Join Example

• **EMPLOYEE**

EMP_NAME	STREET	CITY
Kamran	Civil line	Muzaffarabad
Saqib	Park street	Kel
Haider	M.G. Street	Lahore
Bilawal	View city	Hyderabad

• **FACT_WORKERS**

EMP_NAME	BRANCH	SALARY
Kamran	Infosys	10000
Saqib	Wipro	20000
Ayyaz	HCL	30000
Bilawal	TCS	50000



Right Outer Join Example

- **INPUT:** (EMPLOYEE ⋈ FACT_WORKERS)
- **OUTPUT:**

EMP_NAME	BRANCH	SALARY	STREET	CITY
Kamran	Infosys	10000	Civil line	Muzaffarabad
Saqib	Wipro	20000	Park street	Kel
Bilawal	TCS	50000	View city	Hyderabad
Ayyaz	HCL	30000	NULL	NULL



Full Outer Join

- Full outer join is like a left or right join except that it contains all rows from both tables.
- In the full outer join, ***tuples in R that have no matching tuples in S and tuples in S that have no matching tuples in R in their common attribute name.***
- It is denoted by $\bowtie$.



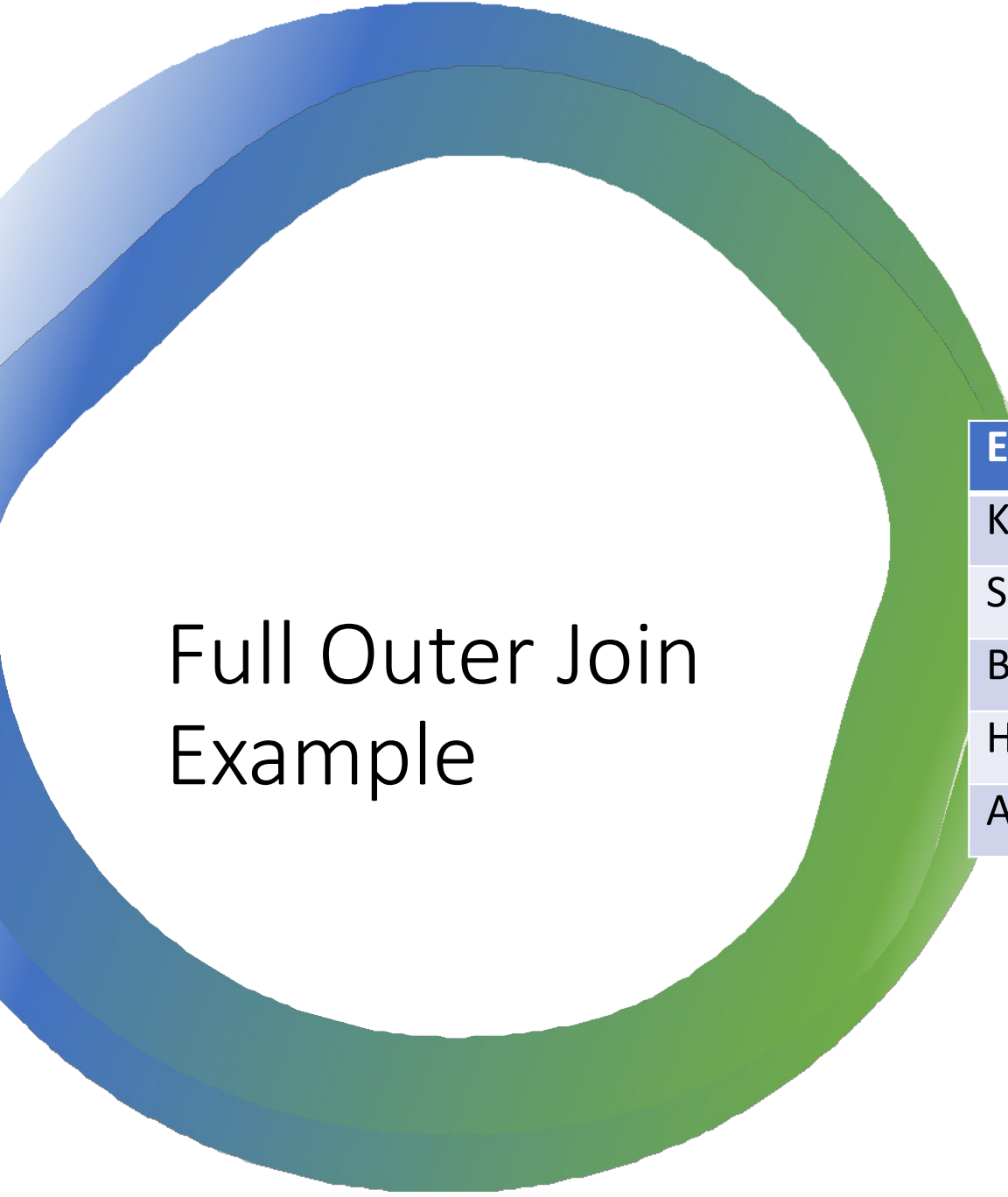
Full Outer Join Example

• EMPLOYEE

EMP_NAME	STREET	CITY
Kamran	Civil line	Muzaffarabad
Saqib	Park street	Kel
Haider	M.G. Street	Lahore
Bilawal	View city	Hyderabad

• FACT_WORKERS

EMP_NAME	BRANCH	SALARY
Kamran	Infosys	10000
Saqib	Wipro	20000
Ayyaz	HCL	30000
Bilawal	TCS	50000



Full Outer Join Example

- **INPUT:** (EMPLOYEE to
FACT_WORKERS)

- **OUTPUT:**

EMP_NAME	STREET	CITY	BRANCH	SALARY
Kamran	Civil line	Muzaffarabad	Infosys	10000
Saqib	Park street	Kel	Wipro	20000
Bilawal	View city	Hyderabad	TCS	50000
Haider	M.G. Street	Lahore	NULL	NULL
Ayyaz	NULL	NULL	HCL	30000



A Venn diagram illustrating an Equi Join. It consists of two overlapping circles. The left circle is blue and the right circle is green. The intersection of the two circles is shaded a darker blue-green. The text 'Equi Join' is centered within the intersection area.

Equi Join

- It is also known as an *inner join*.
- It is the most common join.
- It is based on matched data as per the equality condition.
- The equi join uses the comparison operator(=).



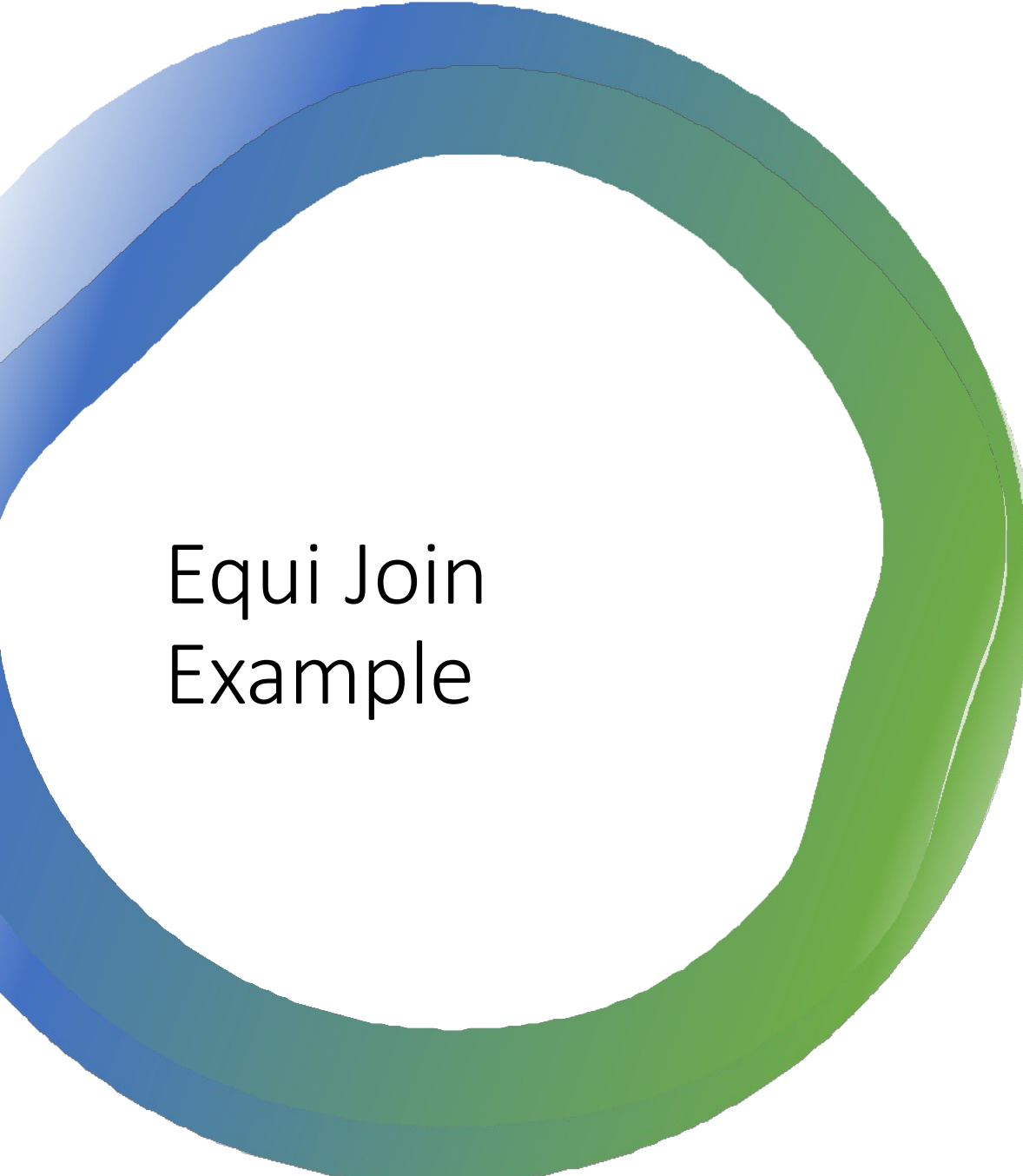
Equi Join
Example

- **CUSTOMER**

CLASS_ID	NAME
1	Jhon
2	Harry
3	Jackson

- **PRODUCT**

PRODUCT_ID	CITY
1	Delhi
2	Mumbai
3	Noida



Equi Join Example

- **INPUT:** (CUSTOMER ⋈ PRODUCT)
- **OUTPUT:**

CLASS_ID	NAME	PRODUCT_ID	CITY
1	Jhon	1	Delhi
2	Harry	2	Mumbai
3	Jackson	3	Noida